

SHKODRA, Albania

WMO: 136000

Lat: 42.100N

Lon: 19.533E

Elev: 43

StdP: 100.81

Time Zone: 1.00 (EUC)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.5	-0.9	-12.7	1.3	2.5	-9.4	1.7	3.6	10.5	5.2	9.9	5.0	1.7	90	0.402

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.0	35.9	22.8	34.4	22.8	33.0	22.5	25.3	31.2	24.5	30.7	23.7	30.0	2.7	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.7	18.7	28.3	22.7	17.6	27.5	21.9	16.7	27.0	77.7	31.0	74.2	30.5	71.3	30.0	33.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
9.7	7.3	5.8	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	-6.2	27.9	2.6	1.7	-8.1	29.2	-9.6	30.2	-11.1	31.1	-13.0	32.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.4	5.8	8.4	11.9	15.5	18.5	23.5	26.7	27.0	22.5	17.2	12.5	7.2
	DBStd	7.73	3.58	3.45	2.77	2.85	2.93	3.33	2.37	2.65	3.05	2.76	3.01	3.11
	HDD10.0	317	133	62	13	1	0	0	0	0	0	0	11	97
	HDD18.3	1569	387	278	199	91	34	2	0	1	3	53	175	346
	CDD10.0	2669	4	17	72	167	264	406	517	527	375	224	86	9
	CDD18.3	879	0	0	0	7	40	158	259	269	128	18	1	0
	CDH23.3	8607	0	1	1	54	243	1458	2838	2958	973	81	0	0
	CDH26.7	3811	0	0	0	6	38	571	1374	1478	341	4	0	0
Wind	WSAvg	2.4	2.3	2.7	2.8	2.3	2.4	2.3	2.3	2.3	2.7	2.3	2.6	2.3
Precipitation	PrecAvg	1670	184	150	153	135	90	56	37	59	148	185	243	231
	PrecMax	2797	432	344	476	381	349	247	130	340	504	473	493	524
	PrecMin	621	9	19	7	22	1	2	0	0	7	0	44	24
	PrecStd	689	123	95	112	102	82	57	38	72	129	135	127	138
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.8	19.7	22.8	27.2	29.7	35.0	37.5	38.7	34.3	27.3	21.7	16.2
		MCWB	12.5	12.2	12.6	17.6	20.1	23.3	23.5	22.6	22.0	19.3	17.0	13.6
	2%	DB	13.7	16.7	20.6	24.8	27.6	33.2	35.6	36.5	32.4	25.5	20.4	14.9
		MCWB	11.0	12.0	12.7	16.0	19.0	22.9	23.2	22.5	21.6	17.9	16.5	12.0
	5%	DB	12.1	14.8	18.7	22.9	26.1	31.6	34.3	34.9	30.4	24.0	18.8	13.5
		MCWB	9.6	11.2	11.7	14.9	18.7	22.1	23.1	22.3	20.7	17.5	14.7	10.2
	10%	DB	10.8	13.4	17.1	21.2	24.3	30.0	32.8	33.2	28.6	22.4	17.3	12.2
		MCWB	8.5	10.1	11.3	14.5	17.9	21.8	22.5	22.4	20.4	17.3	14.0	9.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.2	14.7	15.3	18.8	21.8	26.0	26.6	26.4	24.7	21.4	18.6	14.4
		MCDB	14.8	17.3	19.1	24.5	27.2	31.5	32.4	31.9	28.9	24.2	20.6	15.8
	2%	WB	11.5	13.0	14.2	17.7	20.5	24.7	25.3	25.1	23.5	19.9	16.9	13.0
		MCDB	13.3	15.5	17.9	22.7	25.1	30.3	31.4	31.4	28.1	22.8	19.6	14.5
	5%	WB	10.2	11.8	13.3	16.4	19.6	23.7	24.4	24.3	22.6	19.0	15.8	11.2
		MCDB	11.7	14.3	17.0	21.0	24.2	29.0	31.2	30.7	27.6	22.4	18.3	12.8
	10%	WB	9.0	10.6	12.3	15.4	18.7	22.7	23.6	23.5	21.7	18.1	14.6	9.8
		MCDB	10.5	13.2	15.6	19.7	23.1	28.0	30.5	30.1	27.0	21.2	16.7	11.5
Mean Daily Temperature Range	5% DB	MDBR	7.0	7.5	8.7	9.2	9.0	9.8	11.3	12.0	10.4	9.4	7.8	7.4
		MCDBR	8.2	10.0	12.2	12.4	11.6	11.4	13.2	13.7	12.7	11.7	9.4	8.7
		MCWBR	5.7	6.0	5.9	5.8	5.2	4.7	5.4	4.7	5.0	6.0	5.9	5.9
	5% WB	MCDBR	6.8	8.7	9.8	10.4	10.0	10.4	11.6	11.9	10.8	9.3	7.5	6.0
		MCWBR	5.4	6.0	6.1	5.8	5.5	5.2	5.8	5.5	5.8	6.1	5.6	5.2
Clear-Sky Solar Irradiance	taub	0.342	0.368	0.417	0.472	0.461	0.439	0.441	0.456	0.435	0.401	0.370	0.343	
	taud	2.398	2.305	2.186	2.056	2.156	2.271	2.259	2.209	2.256	2.326	2.362	2.401	
	Ebn at Noon	791	823	821	800	821	839	833	806	795	777	746	753	
	Edh at Noon	81	103	131	160	149	133	133	136	121	100	84	75	
All-Sky Solar Radiation	RadAvg	1.67	2.42	3.58	4.65	5.81	6.73	6.96	6.18	4.43	3.07	1.94	1.48	
	RadStd	0.19	0.36	0.49	0.51	0.46	0.45	0.37	0.39	0.36	0.32	0.20	0.27	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (25 neighbors)	+0.47	N/A	N/A	N/A	N/A	N/A	N/A	-103	N/A	N/A	N/A

Nomenclature: See separate page